AGU PUBLICATIONS

Fractional OAM vortex SAR imaging based on Chirp scaling algorithm

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Key Points

·We derive the echo formula for fractional vortex SAR imaging.

·We simulate the amplitude and phase patterns of plane wave and integral and fractional order vortex wave.

·We have done imaging experiments of multiple targets and single target under Gaussian signal-to-noise ratio.

Abstract

Vortex electromagnetic wave carries orbital angular momentum. Combined with Doppler information provided by radar platform movement, vortex electromagnetic wave can achieve higher resolution target imaging in SAR imaging technology. In this paper, fractional order OAM vortex SAR imaging is studied. Firstly, the side-looking strip SAR imaging model is established. Then, the scattering echo equation of fractional order OAM is derived. Finally, the imaging simulation of multi-point target and single point target under Gaussian SNR is carried out by Chirp Scaling algorithm. The experimental results show that compared with the integer order OAM vortex SAR imaging, the fractional order OAM vortex SAR imaging in this paper has stronger robustness in multi-target and noise environment, which proves the effectiveness of the fractional order vortex SAR imaging.

Plain Language Summary

We verify by chirp scaling algorithm. In this paper, the fractional-order OAM vortex SAR imaging has strong robustness in multi-target and noise environment.

1. Introduction

Vortex electromagnetic waves carrying orbital angular momentum are highly valued by researchers [1]. Due to the orthogonality of OAM in different modes, the vortex EM waves in different modes are independent of each other [2]. Therefore, they have significant potential for application in high-wideband data transmission and are very promising for research in wireless communication [3], electromagnetic imaging [4], SAR imaging [5], and other fields.

In [6], G. Guo from National University of Defense Technology took the lead in applying vortex electromagnetic wave to the field of radar imaging, and pointed out that vortex electromagnetic wave has great potential in target azimuth direction imaging, which provides a new idea for the development of new radar system. In [7], K. Liu derived the echo signal model in multiple transmit multiple receive (MIMO) and multiple transmit single receive (MISO) modes, and obtained the two-dimensional focusing result of the target by fast Fourier Transform (FFT) and BP imaging algorithm. In [8],

H. Liu proposed an electromagnetic vortex enhanced imaging method based on fractional OAM beam to combat the influence of noise, and proposed a method to generate fractional OAM beam with non-integer OAM mode phase distribution by using uniform circular array, and analyzed its characteristics. In [9], based on the side-looking strip vortex SAR imaging model, J. Wang used the RD algorithm to obtain the target imaging results. Theoretical and simulation results show that the side-looking strip vortex SAR technique can more clearly characterize the target in the beam compared with the traditional SAR technique. In [10], H. Zheng reviewed the latest advances in theoretical models, propagation, generation, measurement and application of fractional vortex beams, and considered possible future development directions and challenges.

In recent years, many achievements have been made on integer order OAM vortex SAR imaging. From the current research, it is of great significance for fractional order OAM vortex SAR imaging. Based on this, this paper first derives the imaging echo formula of side-looking strip SAR, and then selects the chirp scaling algorithm to encode the echo. Finally, the simulation experiment of the detection target is carried out. The experimental results show that the fractional-order OAM vortex SAR imaging is more robust than the integer imaging in the multi-target and noise environment, which provides a feasible reference scheme for electromagnetic vortex SAR imaging.

2. Vortex SAR imaging scenario and echo equation

2.1. Imaging scene

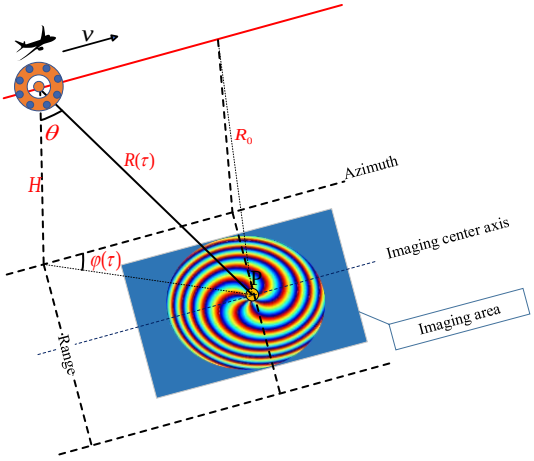


Figure 1. Scenario geometry of side-looking vortex SAR

Electromagnetic vortex synthetic Aperture Radar (SAR) images stationary targets on the ground by means of an aircraft platform [11]. In the process of SAR movement, the SAR pulses are transmitted and received with a certain pulse repetition period. The incident wave is backscattered by the scattering elements of the UCA antenna to form the SAR echo information. Its imaging scene is shown in Figure 1.

$R(\tau)$ is the instantaneous distance between the radar and the target, H is the altitude of the SAR flight, v is the altitude of the SAR flight, Y_c is the imaging central axis, φ(τ) represents the instantaneous azimuth information, and the coordinates of the imaging point target are P (X, Y ,0).

$$\begin{cases} R(\tau) = \sqrt{(X - v\tau)^2 + Y_c^2 + H^2} \\ \varphi(\tau) = \arccos\left(\dfrac{X - v\tau}{\sqrt{R^2(\tau) - H^2}}\right) \end{cases} \tag{1}$$

High resolution in azimuth direction is achieved by forming a long "virtual aperture" by relying on SAR motion, and high resolution in range direction is achieved by transmitting a linear frequency modulation signal with a certain bandwidth by pulse compression. The ideal range dimension and azimuth dimension signal can be expressed as follows [12]:

$$\begin{aligned} S(t,\tau) = {} & \omega_r\left(t - \frac{2R(\tau)}{c}\right)\omega_a(\tau - \tau_c) \\ & \times \exp\left\{j\pi K_r\left(t - \frac{2R(\tau)}{c}\right)^2\right\} \\ & \times \exp\left\{-j\frac{4\pi f_c R(\tau)}{c}\right\} \end{aligned} \tag{2}$$

Where, $\omega_r(\cdot)$ and $\omega_a(\cdot)$ represent the range and azimuth envelope; c is the speed of light; t is the distance time variable; τ is the azimuth time variable; K_r is the rate of modulation; f_c is the center frequency of the antenna.

2.2. Signal echo

In this paper, a uniform circular array of transmit antennas is used to generate vortex electromagnetic waves [13]. From the perspective of mathematics, the fractional phase term can be expressed

as the sum of Fourier series superposition of orthogonal integral states, so the fractional OAM can be approximately composed of finite proportional integral OAM [14], and the phase term carrying fractional orbital angular momentum satisfies [15]:

$$\begin{cases} e^{jl\phi_n} = \sum_{-\infty}^{\infty} \dfrac{e^{jl\pi}\sin l\pi}{\pi(l-\alpha)} e^{j\alpha\phi_n} \\ s_l = \dfrac{e^{jl\pi}\sin l\pi}{\pi} \sum_{\alpha=\alpha_{\min}}^{\alpha_{\max}} \dfrac{j^{-\alpha}e^{j\alpha\phi_n}}{J_\alpha(ka\sin(\theta))(l-\alpha)} \end{cases} \tag{3}$$

Where l is the obtained fractional order, $\alpha_{\min}$ and $\alpha_{\max}$ is the minimum and maximum modes of the integrated OAM beam that can be generated by the UCA. The first kind Bessel function can be expressed as:

$$J_\alpha(ka\sin(\theta)) = \frac{j^{-\alpha}}{2\pi}\int_0^{2\pi} e^{-jka\sin(\theta)\cos\phi-\alpha\phi} d\varphi .$$

The transmitted signal satisfies:

$$\begin{aligned} S_e(t,\tau,l) &= -\frac{j\mu_0\omega}{4\pi}\sum_{n=0}^{N-1} S(t,\tau)\cdot s_l \cdot \int \frac{e^{jk|r-r_n|}}{|r-r_n|} d\tilde{V}_n \\ &\approx -\frac{j\mu_0\omega dN}{4\pi}\frac{e^{jkr}}{r}S(t,\tau) \\ &\times \frac{e^{jl\pi}\sin l\pi}{\pi}\sum_{\alpha=\alpha_{\min}}^{\alpha_{\max}}\frac{e^{j\alpha\varphi}}{(l-\alpha)} \\ &\approx \frac{j\mu_0\omega dN}{4\pi}\frac{e^{jkr}}{r}S(t,\tau)\cdot e^{jl\varphi} \end{aligned} \tag{4}$$

Where, μ_0 is the permeability constant in vacuum; ω is the angular frequency of the electromagnetic wave; $k=\omega/c$ is the wave number; j is the current density at each array element; d is the oscillator length of the array element; $\int(\cdot)d\tilde{V}_n$ represents the integral of the current element on the counter element; The last line expression in Equation (4) is an approximation when the number of array elements N is sufficiently large.

The scattering intensity of a single target point P is denoted as σ, the receiving antenna uses a single antenna to receive [16], since it is received at the center point of UCA, the echo received by the nth array element satisfies:

$$S_r(t,\tau,l) \approx -\frac{j\mu_0\omega d^2 N}{4\pi} S(t,\tau)\cdot\sigma\cdot\frac{e^{j2kr}}{r^2}\cdot e^{jl\varphi} \tag{5}$$

The constant term independent of phase is compensated and omitted, and the echo signal of multi-point target satisfies:

$$S_M(t,\tau,l) = \sum_{m=0}^{M} \left\{ \begin{array}{l} \omega_r(t-\frac{2R(\tau)}{c})\omega_a(\tau-\tau_c) \\ \times\exp\{j\pi K_r(t-\frac{2R(\tau)}{c})^2\} \\ \times\exp\{-j\frac{4\pi R(\tau)}{\lambda}\}\cdot\bar{\sigma}\cdot e^{jl\varphi(\tau)} \end{array} \right\} \tag{6}$$

Where, $\bar{\sigma} = \sigma_m \cdot \dfrac{e^{j2kr_m}}{r_m^2}$, $\lambda = 2\pi / f_c$, $\varphi(\tau)$ is the instantaneous bearing information during flight.

3. Chirp Scaling imaging algorithm

According to the multi-point target echo signal, compared with the traditional SAR echo signal, the vortex SAR echo signal introduces an additional phase term. Compared with the conventional SAR signal, because of the phase dependence of the azimuth term, the vortex signal has a more obvious ability to distinguish the target in the beam than the plane wave signal. In this paper, the vortex SAR imaging algorithm is used, and the imaging process is shown in Figure 2.

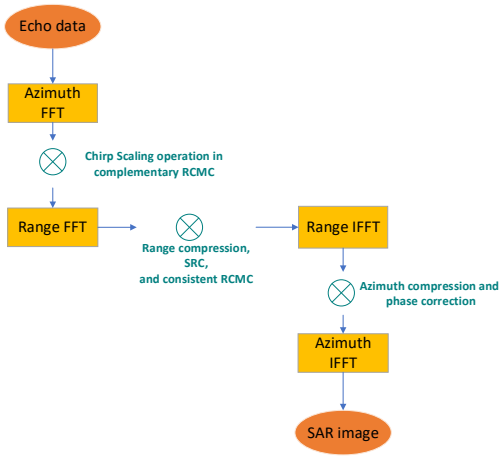


Figure 2. Chirp Scaling Algorithm flowchart.

In the first step, consistent with the traditional CS algorithm [17], Chirp Scaling processing is implemented by multiplying a function $H_1(t, f_a)$ in the range-Doppler domain, and each range cell target RCMC is normalized to align with the reference curvature trajectory.

$$H_1(t, f_a) = \exp\{j\pi K_s(f_a)C_s(f_a)[t-\frac{2R(\tau)}{c}]^2\} \tag{7}$$

In the second step, range compression, quadratic range compression (SRC) and consistent RCMC [18] are achieved by multiplying a function $H_2(f_r, f_a)$ in the two-dimensional frequency domain.

$$H_2(f_r, f_a) = \exp\{j\pi \frac{f_r^2}{K_s(f_a)[1+C_s(f_a)]}\} \times \exp\{j4\pi \frac{R_0 C_s(f_a)}{c/f_r}\} \quad (8)$$

In the third step, a function $H_3(t, f_r)$ is multiplied in the range-Doppler domain to achieve azimuth compression and compensate for the phase residual introduced by Chirp Scaling. According to the above steps, the target range azimuth focusing can be finally achieved.

$$H_3(t, f_r) = \exp\{j2\pi \frac{ct}{\lambda}[1-\sqrt{1-(\frac{\lambda f_a}{2v})^2}]\} \times \exp\{j\pi K_s(f_a)[1+C_s(f_a)](\Delta t)^2\} \quad (9)$$

Where, f_a is the azimuth term Doppler frequency, $C_s(f_a)$ is the range migration factor, $K_s(f_a)$ is the range dimensional chirp rate, and the reference distance $R_0 = \sqrt{Y_c^2 + H^2}$, $r = ct/2$, $\Delta t = 2(R_0 - r)/c$.

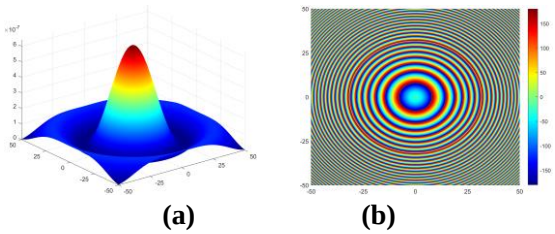
4. Simulation experiment

Table 1: SAR experimental parameters

Parameters	value	Unit
Frequency f_c	40	G Hz
Radar flight speed *v*	300	m / s
Radar flight altitude *H*	10	Km
Antenna center angle β	30	°
Scope of OAM mode *l*	[0,10]	-
Width of pulse T_r	2	μ s
Bandwidth B_r	300	M Hz
Synthetic aperture length	200	m

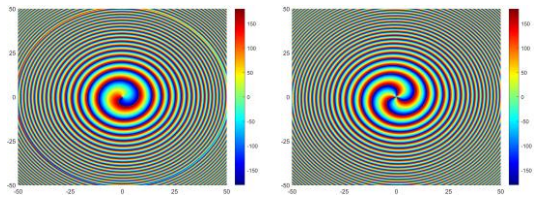
Experiment 1: Vortex wavefront experiment

Due to the effect of integral mode OAM, the fractional order vortex wavefront is very different from the integer order vortex wavefront, and the difference lies in the influence of Bessel function. Therefore, we have done the experimental simulation of fractional order and integer order vortex wavefront, and compared the wavefront amplitude and phase. The experimental parameters are shown in Table 1. Figure 3 (a) and (b) demonstrate the conventional plane wavefront phase and amplitude, respectively. Figures 4 (a), (b) and (c) demonstrate the conventional vortex wavefront phase; Figures 5 (a), (b) and (c) demonstrate the conventional vortex wavefront amplitude; FIG. 6 (a), (b) and (c) demonstrate the fractional vortex wavefront phase of the proposed method. FIG. 7 (a), (b) and (c) demonstrate the fractional vortex wavefront amplitude in this paper.

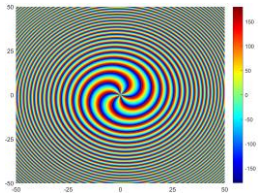


(a) **(b)**

Figure 3. Phase and amplitude images of conventional plane wavefront (a). Phase images (b). Amplitude map



(a) **(b)**



(c)

Figure 4. Phase diagram of conventional vortex wavefront (a). Mode l=1 (b). Mode l=3 (c). Mode l=6

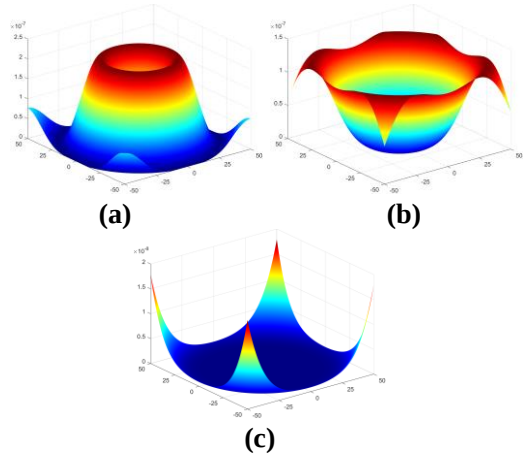


Figure 5. Amplitude diagram of conventional vortex wavefront (a). Mode l=1 (b). Mode l=3 (c). Mode l=6

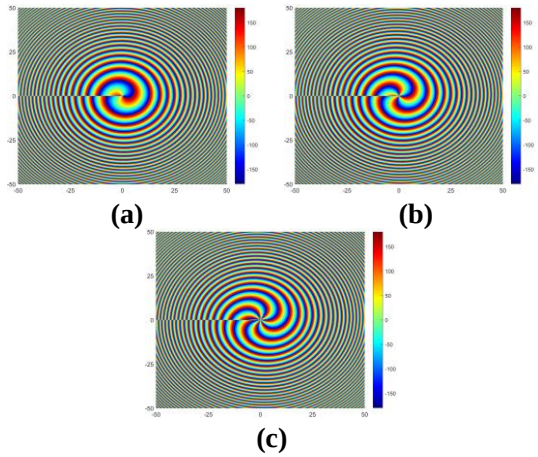


Figure 6. Phase diagram of fractional vortex wavefront. (a). Mode l=1.5 (b). Mode l=3.5 (c). Mode l=6.5

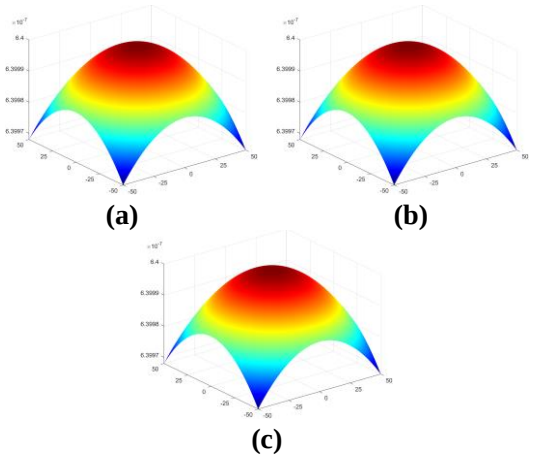


Figure 7. Fractional vortex wavefront amplitude diagram. (a). Mode l=1.5 (b). Mode l=3.5 (c). Mode l=6.5

From Figure 5, we know that with the increase of modes, the divergence of traditional vortex wavefront amplitude becomes stronger and stronger, which is caused by the influence of Bessel function. From Figure 7, we know that with the increase of modes, the amplitude of the fractional vortex wavefront in this paper is relatively stable, which is due to the integral OAM mode, so the effect will be better.

Experiment 2: Multi-point target SAR imaging

In this experiment, 20-point targets were imaged, and the point target model diagram is shown in Figure 8. We compare the imaging effects under different OAM modes with the same experimental parameters through simulation experiments. Figure 9 shows the imaging effects of the same multiple targets in integer mode and fractional mode.

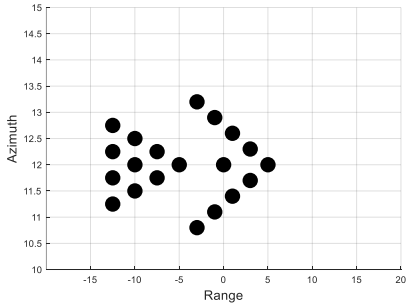


Figure 8. 20-point target.

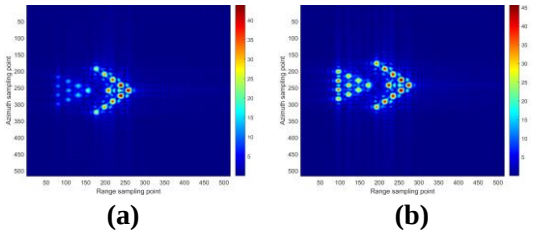


Figure 9. Comparison of imaging results of 20-point targets in different modes (a). Integer order l=6 (b). The fractional order l=6.5

From the imaging section diagrams of different OAM modes in the multi-target imaging scene in Figure 9 (a) and (b), we conclude that the imaging effect of fractional OAM mode is more robust than that of integer OAM mode. This is because the position of each target is very close, and Bessel has a certain influence on the amplitude of echo signal, so the imaging effect of fractional order is better than that of integer order.

Experiment 3: SNR experiment

Noise is inevitable in practical applications, so in this experiment, a certain amount of Gaussian white noise is added to the echo signal to carry out imaging comparison experiments of different modes. We did two sets of SNR experiments, one with SNR of -15 and the other with SNR of -25. The experiment was performed on a single point target with the coordinates: $(-20,\ Y_c\ -10, 0)$. Figure 10 and Figure 11 respectively show the section diagram and azimuth contour of the point target under the same SNR and different modes.

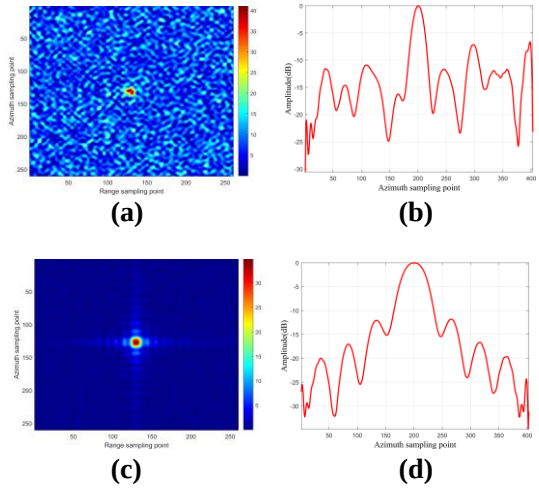


FIG. 10. Section diagrams and azimuth profiles of point targets in different modes when SNR=-15 (a). Mode l=3 point target section diagram (b). Mode l=3 azimuth profile (c). Mode l=3.5 target section (d). Mode l=3.5 azimuth profile

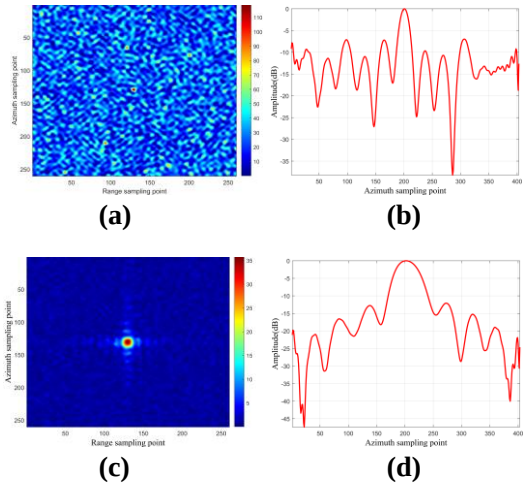


FIG. 11. Section diagrams and azimuth profiles of point targets in different modes when SNR=-25 (a). Mode l=3 point target section diagram (b). Mode l=3 azimuth profile (c). Mode l=3.5 azimuth profile

From the imaging results in Figure 10 and Figure 11, we know that the imaging effect of fractional OAM mode is better than that of integer OAM mode under different SNR. From Figs. 10 and 11, we can also conclude that SNR will affect the imaging effect of point targets. The effect of fractional mode SAR imaging is better in noisy environment, because the fractional order OAM mode can make full use of the number of OAM modes compared with the integer order OAM mode.

5. Conclusions

In this paper, the imaging geometry model and echo signal model of fractional order OAM vortex SAR are established. Based on the Chirp Scaling imaging algorithm, point target simulation experiments verify the effectiveness of the fractional order OAM vortex SAR algorithm. In the multi-target imaging scene, the imaging section of different OAM modes can be seen that the imaging effect of fractional order is more robust than that of integer order. This is because the integer order vortex SAR imaging echo contains the first Bessel function, Bessel will affect the imaging performance of point targets to a certain extent. Theoretical and simulation results show that fractional order vortex SAR imaging is very robust in noisy environment. As the SNR increases, the imaging performance becomes worse, which is the result of the noise signal severely interfering with the imaging signal. In addition, fractional OAM can make full use of the number of OAM modes. Therefore, compared with the conventional vortex SAR imaging method, the vortex SAR imaging method of the proposed fractional OAM is more robust.

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Data Availability Statement

The data files used in this paper are available at (ly,2024).

References

LY/OAM-CS: OAM-CS (OAM-CS). (2024). [Dataset]. https://github.com/imustyl/ly

[1] Liu, D., Shi, H., Yang, T., & Qiao, Z. (2024). Enhanced Resolution Method for Electromagnetic Vortex Imaging Based on Electromagnetic Information Theory. IEEE Transactions on Geoscience and Remote Sensing. https://doi.org/10.1109/TGRS.2024.3423424

[2] Jia, Y., Hu, Q., Li, Z., & Wang, H. (2024). High resolution imaging of acoustic orbital angular momentum wave based on orthogonal matching tracking. *The Journal of the Acoustical Society of America*, *155*(2), 1240-1252. https://doi.org/10.1121/10.0024768

[3] Yang, H., Zheng, S., Zhang, H., Li, N., Shen, D., He, T., . . . Yu, X. (2023). A THz-OAM wireless communication system based on transmissive metasurface. *IEEE Transactions on Antennas and Propagation*, *71*(5), 4194-4203. https://doi.org/10.1109/TAP.2023.3255539

[4] Guo, S., He, Z., Fan, Z., & Chen, R. (2020). CUCA based equivalent fractional order OAM mode for electromagnetic vortex imaging. *IEEE Access*, *8*, 91070-91075. https://doi.org/10.1109/ACCESS.2020.2995149

[5] Zhang, B., Xu, G., Zhou, R., Zhang, H., & Hong, W. (2022). Multi-channel back-projection algorithm for mmWave automotive MIMO SAR imaging with Doppler-division multiplexing. *IEEE Journal of Selected Topics in Signal Processing*, *17*(2), 445-457. https://doi.org/10.1109/JSTSP.2022.3207902

[6] Guo, G., Hu, W., & Du, X. (2013). Electromagnetic vortex based radar target imaging. *J. Nat. Univ. Defense Technol*, *35*(6), 71-76. http://en.cnki.com.cn/Article_en/CJFDTOTAL-GFKJ201306013.htm

[7] Liu, K., Cheng, Y., Yang, Z., Wang, H., Qin, Y., & Li, X. (2014). Orbital-angular-momentum-based electromagnetic vortex imaging. *IEEE Antennas and Wireless Propagation Letters*, *14*, 711-714. https://doi.org/10.1109/LAWP.2014.2376970

[8] Liu, H., Wang, Y., Wang, J., Liu, K., & Wang, H. (2021). Electromagnetic vortex enhanced imaging using fractional OAM beams. *IEEE Antennas and Wireless Propagation Letters*, *20*(6), 948-952. https://doi.org/10.1109/LAWP.2021.3067914

[9] Wang, J., Liu, K., & Wang, H. (2019). *Side-looking stripmap SAR based on vortex electromagnetic waves.* Paper presented at the 2019 IEEE International Conference on Communications Workshops (ICC Workshops).https://doi.org/10.1109/ICCW.2019.8757161

[10] Zhang, H., Zeng, J., Lu, X., Wang, Z., Zhao, C., & Cai, Y. (2022). Review on fractional vortex beam. *Nanophotonics*, *11*(2), 241-273. https://doi.org/10.1515/nanoph-2021-0616

[11] DU L., WANG, Zhao., WANG, Y., et al. Survey of research progress on target detection and discrimination of single-channel SAR images for complex scenes[J]. Journal of Radars, 2020, 9(1): 34–54.http://dx.doi.org/10.12000/JR19104

[12] Leng, X., Ji, K., & Kuang, G. (2023). Ship detection from raw SAR echo data. *IEEE Transactions on Geoscience and Remote Sensing*, *61*, 1-11. https://doi.org/10.1109/TGRS.2023.3271905

[13] Lv, K., Zhu, G., Ma, H., Liu, H., Shao, S., & Wei, J. (2021). *Three-dimensional imaging using the electromagnetic vortex synthetic aperture radar.* Paper presented at the 2021 CIE International Conference on Radar (Radar). https://doi.org/10.1109/Radar53847.2021.10028448

[14] Liu, K., Cheng, Y., Li, X., Wang, H., Qin, Y., & Jiang, Y. (2016). Study on the theory and method of vortex-electromagnetic-wave-based radar imaging. *IET Microwaves, Antennas & Propagation*, *10*(9), 961-968. https://doi.org/10.1049/iet-map.2015.0842

[15] Kotlyar, V., Kovalev, A., Nalimov, A., & Porfirev, A. (2020). Evolution of an optical vortex with an initial fractional topological charge. *Physical Review A*, *102*(2), 023516. https://doi.org/10.1103/PhysRevA.102.023516

[16] Han, Y., Li, C., Du, Y., (2023). SAR imaging based on OAM-ωK algorithm. *Radio Science*, *58*(7), 1-12.https://doi.org/10.1029/2022RS007613

[17] Yang, T., Shi, H., & Qiao, Z. (2022). Enhanced SAR imaging using multiple plane spiral orbital angular momentum beams. *Measurement*, *202*, 111784. https://doi.org/10.1016/j.measurement.2022.111784

[18] Zhong, H., Huang, J., Wang, M., Wu, H., Ding, Y., Nai, H., . . . Zhang, S. (2022). Focusing Highly Squinted SAR Data of a Short-Range Wide-Swath Region Using Azimuth-Dependent High-Order RCMC and Frequency Extended Nonlinear Chirp Scaling Based on Equidistant Sphere Model. *IEEE Access*, *10*, 129465-129478. https://doi.org/10.1109/ACCESS.2022.3228330